

CSE 400: Fundamentals of Probability in Computing
Project Scribe Submission

Group No.: S1 G17

Domain: ITS

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Scribe Question 1: System Objective and Deterministic Baseline

Clearly restate your project objective. Identify and describe the deterministic baseline strategy you are implementing. Explain why this baseline is reasonable or commonly used in your domain.

Answer:

Our project aims to model traffic flow at a four-way intersection with stochastic traffic arrivals and the deterministic traffic signal controllers. The objective is to quantify congestion and delay in terms of queue length and waiting time of the vehicle.

The arrival of traffic at the intersections are unpredictable, as traffic vehicles arrive at random, and the demand of traffic is not constant all day. The resulting variability causes the variation in queue lengths and congestion levels.

In case of Milestone 3, we apply deterministic baseline strategies that run on definite rules.

The initial one is a **Fixed Timer Controller**. In this controller the traffic light is of constant length cycle in which individual directions are allotted a given duration of green time and a duration of red time to the other directions. The cycle repeats itself regardless of the conditions of the traffic.

The second proposal is a **Time-of-Day (ToD) Controller**. This controller will be operating on varying schedules of signal timing at various times of the day at times like off peak hours, morning peak hours, mid day, evening peak hours and night. The time periods have preset time schedules that are expected to be followed depending on the traffic demand.

The two strategies are deterministic in the sense that the timings of the signals are not determined by real-time traffic measurements. These are the strategies that are usually applied in the actual traffic systems as the strategies are simple, predictable and not hard to implement.

Scribe Question 2: Random Variables in the Simulation

Identify the key random variables in your system. For each variable provide a formal definition, specify the assumed distribution, and explain how it is used within the deterministic baseline simulation. **Answer:**

The primary uncertainty factor of the system is the arrival of vehicles to the intersection. Probabilistic random variables are applied to model vehicle arrivals.

Vehicle Arrivals (N_k)

Where N_k is the number of cars entering the system at time interval Δt . The Poisson distribution models for vehicle arrivals as:

$$N_k \sim \text{Poisson}(\lambda(t)\Delta t)$$

$\lambda(t)$ is the rate of arrival at time t .

Arrival Rate ($\lambda(t)$)

The rate of arrival $\lambda(t)$ indicates the number of vehicles that are expected to arrive in a unit time. This rate in the simulation is found to vary with various times of the day.

Queue Length ($Q(k)$)

There is queue length, which denotes the queue associated with the intersection in time step k . The queue evolves as:

$$Q(k+1) = Q(k) + N_k - D_k$$

D_k is the number of vehicles that leave at the time step.

Waiting Time (W)

Waiting time is the time that vehicles spend waiting to cross over the intersection. The cumulative waiting time of all the vehicles that are present in the system is recorded in the simulation.

In spite of random arrivals and queue states, the signal control policy is deterministic.

Scribe Question 3: Probabilistic Modeling and Fixed Decision Rule

Explain how uncertainty is modeled in your simulation while keeping the decision strategy deterministic.

Answer:

Answer:

Stochasticness of the arrival of vehicles creates uncertainty in the system. The arrival rate in each time interval is of a Poisson distribution:

$$N_k \sim \text{Poisson}(\lambda(t)\Delta t)$$

where the $\lambda(t)$ is time varying to indicate the varying intensities of traffic in the day.

The traffic signal controller is also deterministic although the time when vehicles appear is random.

In the case of the Fixed Timer controller, the signal phase is set to be defined based on cycle which repeats:

$$u(k) = f(k \bmod C)$$

C is the cycle length of the signal.

For Time-of-Day controller case, timing of the signal follows fixed time intervals:

$$u(k) = f(k, \text{time period})$$

In the two strategies, the timing of the signal is not modified depending on the observed queue lengths. This helps the deterministic baseline to play a reference role in assessing higher order adaptive controllers.

Scribe Question 4: Simulation Pipeline

Describe the complete simulation pipeline including input generation, processing stages, deterministic decision mechanism, and output metrics.

Answer: simulating is a four way centered intersection in which the occurrence of vehicles is stochastic. A Non-Homogeneous Poisson Process is used to model the demand in traffic with an equation where the arrival rate is different during the various time periods including off-peak, build-up and peak hours. This enables the simulation to receive the intrinsic randomness appears in the actual city traffic. The application combines Python codes and sumo traffic simulator using the TraCI API. SUMO takes care of the physical motion of the vehicles and Python controls the traffic signal logic, which also collects the real-time statistics used in simulation.

The system workflow:

- Stochastic vehicle arrivals are generated using time-varying Poisson rates.
- SUMO simulates vehicle intersections and movement.
- TraCI takes real-time data like queue length and waiting time.
- There are Deterministic signal-controllers which are Fixed Timer and Time-of-Day to determined the duration of traffic signal phases.
- Queue and delay statistics are recorded for further analyzing.

The following screenshot shows the implementation of the core of the Time-of-Day. Controller and logic that are applied in measuring queue length and vehicle waiting time using TRaCI.

```
def get_green_duration(sim_time):
    """Return green duration (seconds) based on time-of-day demand phase."""
    if sim_time < 50:
        return 15 # Off-peak
    elif sim_time < 150:
        return 25 # Build-up
    else:
        return 40 # Peak hour
```

```
green_duration = get_green_duration(t)

if not in_yellow:
    if phase_timer >= green_duration:
        traci.trafficlight.setPhase(TL_ID, YELLOW_PHASES[current_phase_index])
        in_yellow = True
        phase_timer = 0
else:
    if phase_timer >= YELLOW_DURATION:
        current_phase_index = (current_phase_index + 1) % 4
        traci.trafficlight.setPhase(TL_ID, GREEN_PHASES[current_phase_index])
        in_yellow = False
        phase_timer = 0
```

```
queues = {d: traci.lanearea.getLastStepVehicleNumber(d) for d in DETECTORS}
total_q = sum(queues.values())

vehicle_ids = traci.vehicle.getIDList()
total_wait = sum(traci.vehicle.getWaitingTime(v) for v in vehicle_ids)
```

It is a deterministic strategy as it works only depends on predetermined time schedules and does not confirm to the real queue situation which is observed in the pullover. This realization hence makes a point of reference to analyze the performance shortfalls of deterministic signal control under stochastic traffic demand.

Scribe Question 5: Simulation Outputs and Interpretation

Present and interpret the results obtained from your deterministic baseline simulation.

Answer:

- **Traffic demand generation**

Vehicle arrival rates are generated using a non-homogeneous Poisson model to describe the traffic arrivals. The arrival rate $\lambda(t)$ depends on time of the day phases and is dominated by phases as Off-Peak, Build-Up and Peak times which the simulation is capable of capturing true to life traffic jams and fluctuations.

- **Observed simulation behavior**

The findings indicate that the two controllers have quite small queues and waiting times during low-demand times. With peak demand phases however, there are large spikes in the length of queue and waiting time.

- **Deterministic baseline controllers**

Two strategies in the control of baseline signals are considered, firstly Fixed-Timer controller that has a constant phase time periods and secondly, Time-of-Day (ToD) controller that modulates green signal period depending on demand stages set.

- **Performance metrics**

Detector data are computed to calculate the total queue length at each step of the simulation and cumulative time of vehicle waiting in all lanes. We will consider these metrics for the performance of the intersection in the stochastic traffic demand.

- **Interpretation of results**

The simulation findings indicate that the length of queues and waiting time is very high during peak time. It has been shown that the variability cannot be effectively dealt with using deterministic timing strategies that are determined by the average demand. Also, the congestion accumulates the traffic signals when the vehicle arrival rate exceeds the fixed capacity of intersection.

Scribe Question 6: Technical Gaps and Preparation for Next Stage

Identify current limitations of the deterministic baseline and explain how future milestones will address them.

Answer: There is some probabilistic and modeling limitations in current simulation framework:

- **Arrival Model Assumptions**

Time varying traffic is approximated by the Non-Homogeneous Poisson process but demand may show more clustering and associations. Deterministic Limitations of the Controller to guaranty a degree of precision, certain defects were eliminated during the design of both controllers. In order to ensure a certain level of accuracy, both controllers were designed with some defects removed.

- **Deterministic Controller Limitations**

Both the Fixed-Timer and Time-of-Day controllers only work on fixed schedules and in no way react to the actual conditions of the queues

- **Sensor Measurement Noise**

Detector data obtained through TraCI may contain inaccuracies due to sensor limitations such as vehicle occlusion or closely spaced vehicles.

- **Congestion Threshold Definition**

Congestion theory describes how various temporal intervals or total distance traced by a moving object cause congestion to be experienced at that point. Establishing a suitable level of extreme congestion. events is an unresolved modeling problem.

These limitations can be overmined by answering the following question, which will be the next step of the project. A Markov Marking Decision Process (MDP) controller will be created to be adaptive make signal phase decisions dynamically that optimize observed queue states.

End of Submission